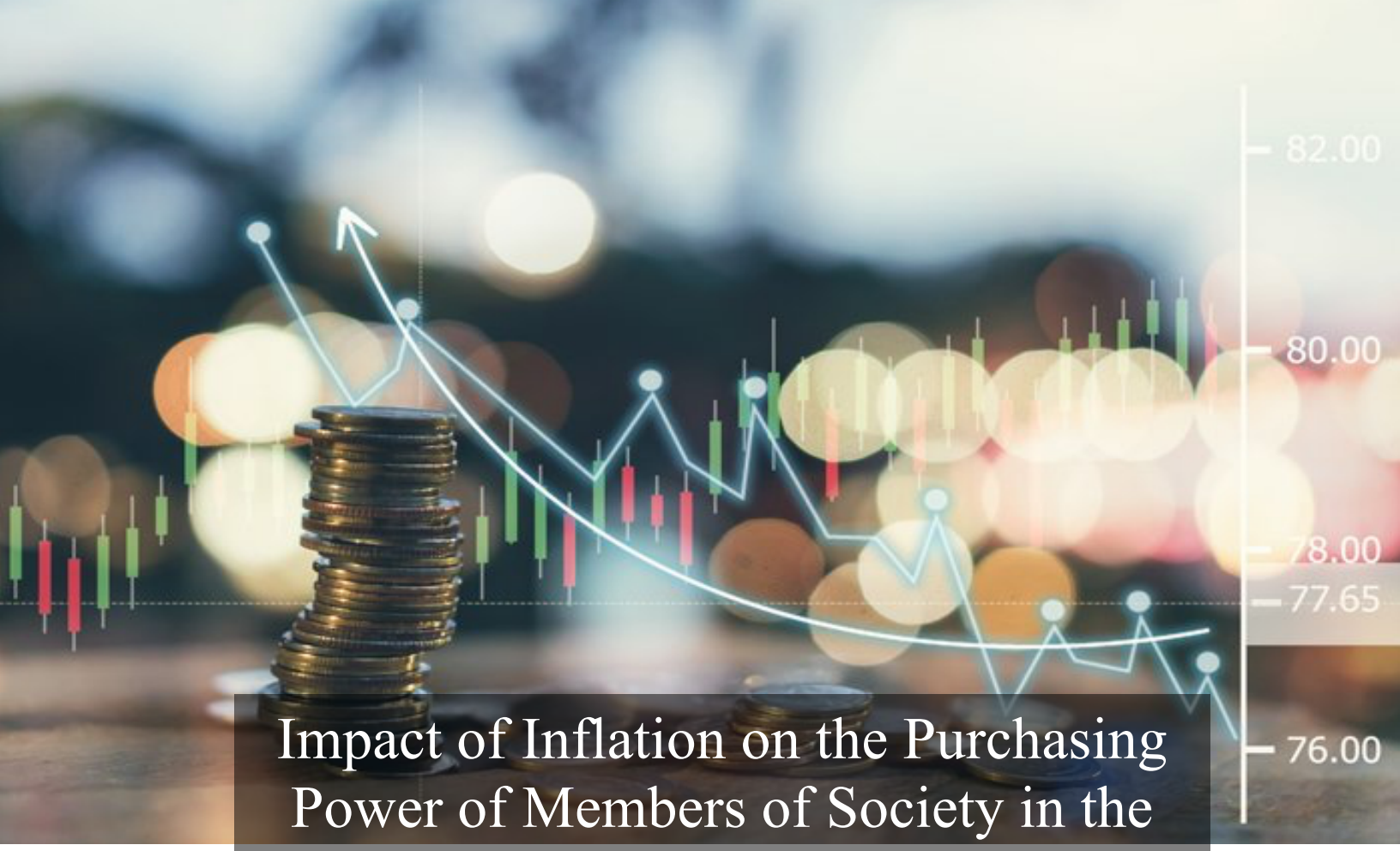

Impact of Inflation on the Purchasing Power of Members of Society in the Kingdom of Saudi Arabia

STAT 238- Introduction to Statistics & Probability Theory Academic Year (2024-2025) – First Semester

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ABSTRACT

Inflation has a profound impact on the economic stability and purchasing power of individuals, influencing their ability to afford goods and services. This study looks at the relationship between inflation and purchasing power across different socioeconomic groups in the Kingdom of Saudi Arabia. Using data from the Saudi Arabian General Authority for Statistics, descriptive analysis, regression models, and chi-square tests are applied to explore these effects. Key findings indicate that lower-income families are more severely burdened by inflation, which in this study greatly altered purchasing and economic habits. Whereas the increase of inflation had less impact on higher-income individuals, the financial burdens in low-income groups were evidenced with specific changes such as dipping into savings and cutbacks on discretionary spending. Regarding conservative spending, a negative albeit weak correlation was established relating the number of dependants with vital expenses. Although significant patterns were identified, the chi-square test failed to establish a strong association of increased living costs with changes in purchase habits, thus suggesting that external factors influence consumer behavior. These findings have important implications for policymakers to develop targeted strategies to mitigate the effects of inflation, particularly on vulnerable segments of the population. This includes increasing the sample size in future research, integrating additional variables such as education and job stability, and accounting for inflation's impact on key sectors like health and education. The study contributes to a better understanding of the socio-economic dynamics in Saudi Arabia and provides a basis upon which further research can build, especially on how best to mitigate inflationary pressures during rapid economic evolution.

INTRODUCTION

Inflation, or the sustained increase in the general price level of goods and services, is the major problem facing economies all over the world. This inflation has wide ramifications, more so in reducing purchasing power, hence constraining an individual's ability to maintain a certain standard of living. Historically, in the Kingdom of Saudi Arabia, the oil-fueled economy and one that has typically represented relatively stable rates of inflation now finds itself, through recent shifts in the world's economy and local policy reforms under the auspices of Vision 2030, exposed to severe tests of inflationary pressures. Changes come daily into the lives of Saudi citizens, influencing how they spend money, save it, or invest it. Historically, Saudi Arabia's dependence on oil revenues insulated it from global economic volatility, and its inflation rates were stable. However, the latest figures indicate a rise in the prices of basic commodities, particularly in the housing and food sectors, which together account for nearly 40% of the CPI basket. Such trends point to a dire need to tackle the unequal effects of inflation on the most vulnerable groups, especially low- and middle-income classes. While higher-income classes may be resilient to such economic pressures, lower-income households often have to make major adjustments in their spending habits and financial priorities to cope with rising costs. This research incorporates data sourced from the Saudi General Authority for Statistics (GASTAT) to investigate the complex nature of the impact of inflation on the purchasing power of individuals in Saudi Arabia. It addresses the inflation challenge from an economic point of view and further ascertains its impact on the stratification of different socioeconomic groups. Using statistical tools in analyzing economic indicators, the study hopes to derive actionable recommendations for policymakers and stakeholders. The idea is to make sure that financial resiliency is ensured and the negative impacts of inflation upon Saudi society are at a minimum so that the goals set under Vision 2030 are not defeated by rising costs and economic inequality.

Methodology

To measure the impact of inflation on the purchasing power of individuals in Saudi Arabia, this study used data and information provided by the Saudi Arabian General Authority for Statistics. the analysis was based on existing statistical datasets that examine economic indicators such as inflation rates , Average Monthly Household Income and so on... . This data was crucial in understanding how inflation affects the affordability of goods and services for different segments of the population in Saudi Arabia. Descriptive statistics, including numerical summaries and graphs, were generated using Minitab software. Linear regression models and statistical tests were applied to evaluate the relationship between (explanatory variable) and (response variable).

The Questionnaire

A total of 12 questions were developed to analyze various aspects and achieve the study's objectives. The questions included in the survey are classified into THREE categories: demographic information, monthly income and living expenses, purchasing habits and price increases, and the impact of price changes on education, health, and living standards.

Part 1: General Information

1. **What is your job type?**
    a. Government
    b. Private sector
    c. Self-employed
    d. Unemployed
2. **How many dependents do you have?**
    a. Short answer
3. **How many family members do you have?**
    a. Short answer

Part 2: Economic Impacts

4. **What is the impact of price increases on your purchasing power?**
    a. Not affected
    b. Slightly affected
    c. Moderately affected
    d. Significantly affected
5. **What is the increase in product prices?**
    a. No increase

   b. Less than 500 SAR
   c. Between 500 and 1500 SAR
   d. More than 1500 SAR
6. **What is the change in living costs?**
   a. No change
   b. Less than 10%
   c. Between 10%-20%
   d. More than 20%
7. **What change have you noticed in your purchasing habits?**
   a. No change
   b. Slight change
   c. Moderate change
   d. Significant change

Part 3: Expenses

8. **What is your monthly income (in SAR)?**
   a. Short answer(number)
9. **What are your essential monthly expenses (in SAR)?**
   a. Short answer(number)
10. **What are your healthcare expenses (in SAR)?**
    a. Short answer(number)
11. **What are your fuel or transport expenses (in SAR)?**
    a. Short answer(number)
12. **Have you used your savings to cover expenses?**
    a. Did not use savings
    b. Less than 5,000 SAR
    c. Between 5,000 and 10,000 SAR
    d. More than 10,000 SAR

The Variables

Variable	Type (Quantitative/Categorical)	Values
Inflation Rate	Quantitative	Monthly inflation (2016-2024)
Changes in prices of basic products	Quantitative	Changes in prices of basic products (2016 -2024)
Impact on Purchasing Power	Categorical	• Not affected, • Slightly • affected, • Moderately affected, Significantly affected
Monthly Income (SAR)	Quantitative	Entered amount
Change in Purchasing Habits	Categorical	• No change • Slight change • Moderate change • Significant change
Healthcare Expenses (SAR)	Quantitative	Entered amount
Essential Expenses (SAR)	Quantitative	Entered amount
Job Type	Categorical	• Government, • Private, • Self-Employed, • Unemployed
Savings Used	Categorical	• Did not use savings, • Less than 5,000 SAR, • 5,000-10,000 SAR, • More than 10,000 SAR
Fuel/Transport Expenses (SAR)	Quantitative	Entered amount

Product Price Increase(GASTAT)	Quantitative	Monthly 2016 -2024
Number of Dependents	Quantitative	Entered number
Number of Family Members	Quantitative	Entered number
Product Price Increase	Categorical	• No increase, • Less than 500 SAR, • 500-1500 SAR, • More than 1500 SAR
Increase in Living Costs	Categorical	• No increase, • Less than 10%, • 10-20%, • More than 20%

Table 1:The variables included in this study

Analysis Methods

A. Variable Analysis:

Variable	**Graph Type**
Inflation Rate(GASTAT)	Time plot
Changes in prices of basic products (GASTAT)	Time plot
Monthly Income (SAR)	Histogram
Fuel/Transport Expenses (SAR)	Histogram
Savings Used (SAR)	Pie graph
Product Price Increase	Bar graph
Product Price Increase(GASTAT)	Time plot
Impact on Purchasing Power	Pie graph

Table 2 :Variable analysis table

B. Relation Analysis: Explanatory variables Vs. Response variables

Explanatory Variable	Response Variable	Type of Relationship
Monthly Income (SAR)	Impact on Purchasing Power	Side-by-Side Boxplot
Job Type	Monthly Income (SAR)	Side-by-Side Boxplot
Increase in Living Costs	Change in Purchasing Habits	Two-Way Table
Number of Dependents	Essential Expenses (SAR)	Scatterplot

Table 3:Relation analysis

1. Variable Analysis

This section will discuss the variables of interests in our study using visual summaries, analyzing them to have a deeper insight of the participants' behavior and attitudes.

1.0 . Inflation Rate(GASTAT)

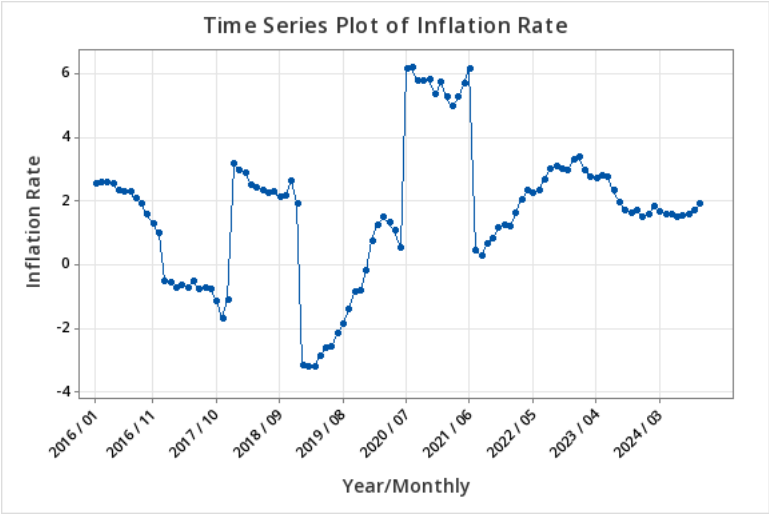

Figure 0 : A Time plot showing the distribution of monthly Inflation Rate(GASTAT)

The time series plot indicates the monthly inflation rate in Saudi Arabia from 2016 to 2024, showing several fluctuations in inflation over recent years. From 2016 to 2018, the inflation rate was stable at around zero, with some periods recording negative rates (deflation), which is indicative of relatively stable price levels. Starting from the year 2018, there was a significant jump in inflation, probably related to economic measures such as the introduction of value-added tax (VAT). In 2020, inflation peaked dramatically, which can be attributed to the global effects of the COVID-19 pandemic and rising global costs. From mid-2021 to 2024, the inflation rate stabilized with minor fluctuations, indicating efforts to achieve economic balance.

1.1 Changes in prices of basic products (GASTAT)

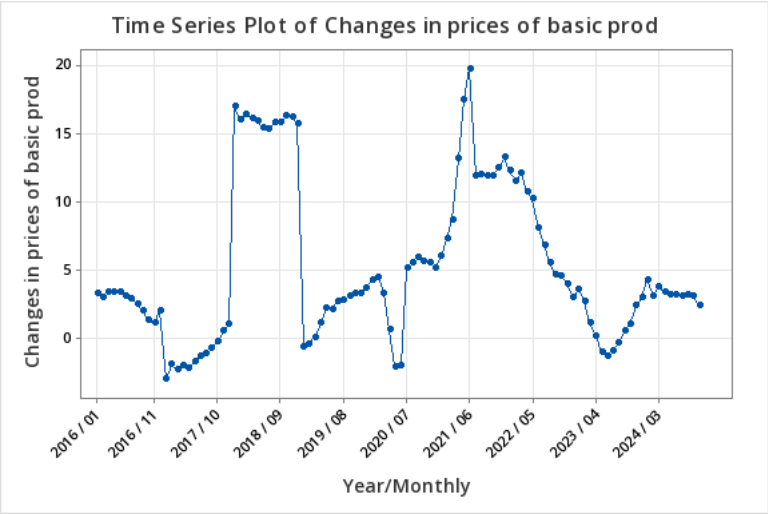

Figure 1 : A Time plot showing the distribution of monthly Changes in prices of basic products ***(GASTAT)***

This time series plot shows the monthly changes in the prices of basic products in Saudi Arabia from 2016 to 2024. In the early period, which ranges from 2016 to 2018, the changes in the prices of basic products were relatively stable and did not show many fluctuations. However, starting from the end of 2018, an upward trend is observed, probably due to economic adjustments like tax reforms or global price pressures. In 2020, there is another significant spike, possibly due to the COVID-19 pandemic disrupting supply chains and global markets. After that, from mid-2021 to 2022, there is heightened price volatility, with some noticeable peaks and troughs. From 2023 onwards, the changes in prices seem to stabilize, at least at somewhat higher levels than in previous years. This pattern reflects ongoing economic adjustments and possible efforts to control inflation in essential goods.

1.2 Monthly Income (SAR)

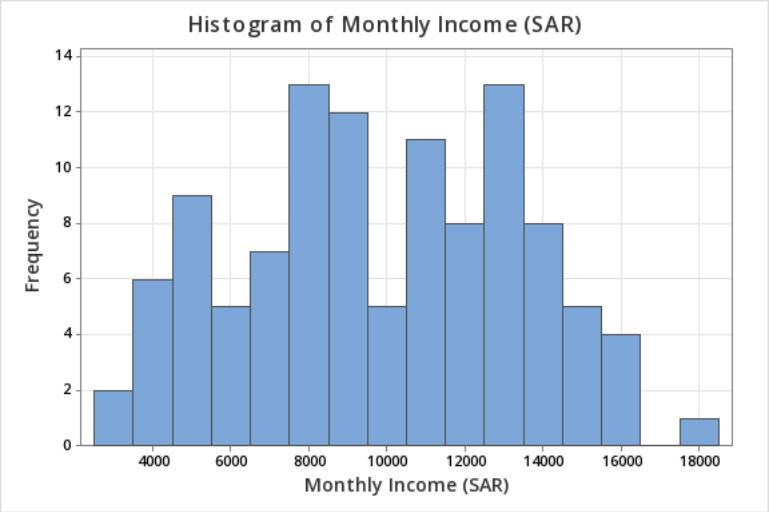

Figure 2 : A histogram showing the distribution of monthly income in SAR

From the histogram of monthly income in SAR, it can be seen that the data is approximately symmetrically distributed with a decided peak in the range of 8,000 to 12,000 SAR. Most of the respondents are within middle-income brackets, with only a few earning below 6,000 SAR and above 15,000 SAR. The mean monthly income is 9,830 SAR, while the median is slightly lower at 9,635 SAR, which indicates a relatively balanced distribution of income among respondents. This would mean that most participants have a moderate income level, with only a small proportion at the lower or higher extremes.

1.3 Fuel/Transport Expenses (SAR

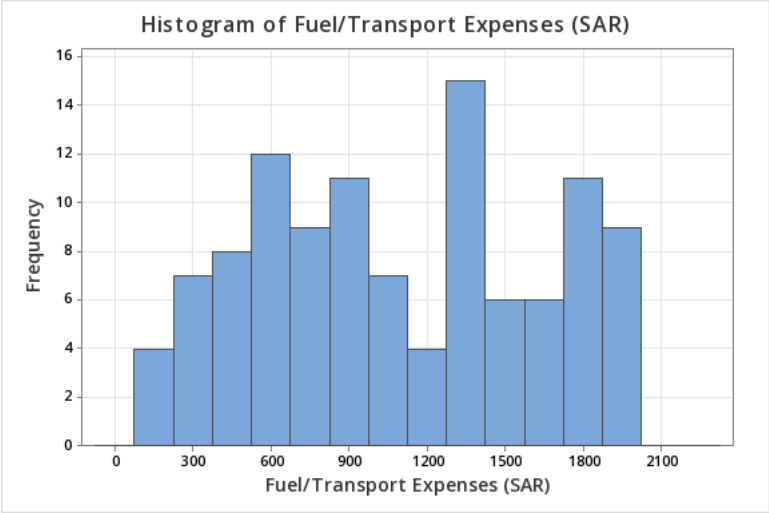

Figure 3 : A histogram showing the distribution of Fuel/Transport Expenses (SAR)

The histogram of Fuel/Transport Expenses (SAR) shows a right-skewed distribution, with the majority of respondents spending between 600 and 1,000 SAR. The most frequent expense range is approximately 800 SAR, which represents the mode of the distribution. The median is likely within the range of 800-1,000 SAR, while the mean is expected to be slightly higher due to the presence of a few higher outliers, reflecting respondents with expenses exceeding 2,000 SAR. The range of expenses varies from around 200 SAR to above 2,000 SAR, indicating moderate variability in transportation costs. This suggests that while most participants have moderate expenses, a small proportion experiences significantly higher transport-related costs.

1.4 Savings Used (SAR)

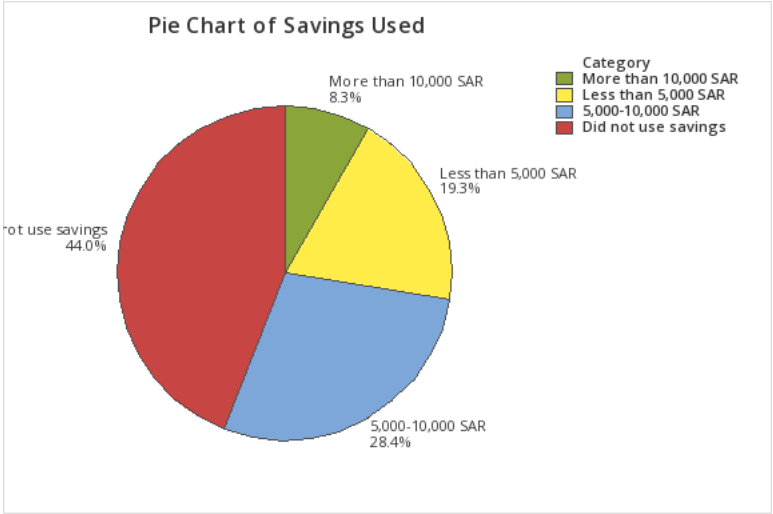

Figure 4: A pie chart showing the distribution of Savings Used (SAR)

The pie chart of Savings Used shows that 44% did not use any savings, meaning that the majority of the participants were in a good position with their finances. On the other hand, 28.4% used savings between 5,000 and 10,000 SAR, reflecting moderate financial strain, while 19.3% used less than 5,000 SAR, suggesting minimal financial impact. Only 8.3% reported using more than 10,000 SAR, showing only a small group faced severe financial challenges. This distribution shows that, while most avoided dipping into their savings, a sizable portion did face various levels of financial strain, although very few needed to use significant amounts of savings.

1.5. Product Price Increase and Product Price Increase(GASTAT)

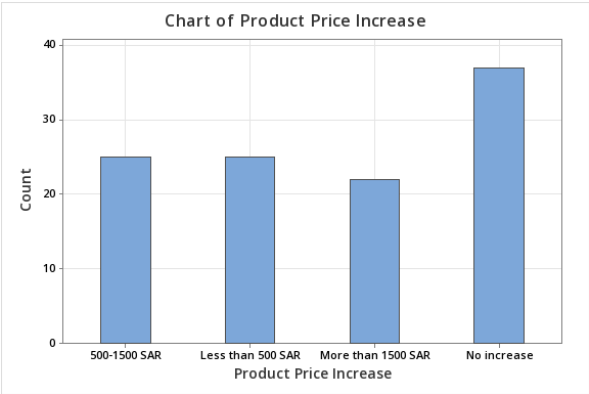

*Figure 5: **A bar chart showing** Product Price Increase*

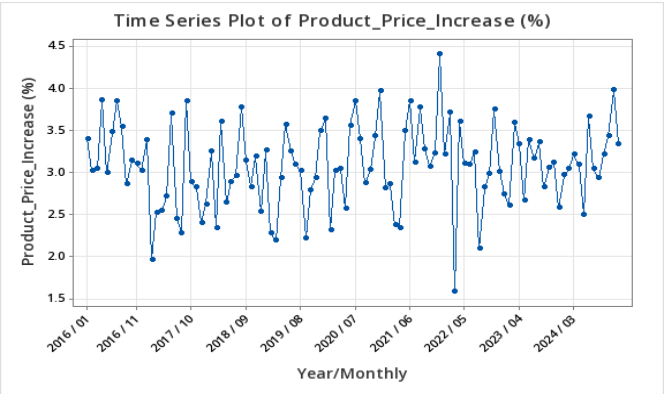

Figure 6: Time plot showing Product Price Increase(GASTAT)

The first chart (bar graph) reflects people's perceptions of price increases, showing that the majority believe there were no price increases ("No increase"), while the other categories, such as increases of less than 500 SAR or more than 1500 SAR, are relatively evenly distributed. On the other hand, the second chart (time series) presents actual data from the General Authority for Statistics, illustrating monthly changes in product price increases from 2016 to 2024. The chart highlights significant fluctuations, with notable peaks, such as in mid-2022. This discrepancy between public perception and official data suggests that people's responses might be influenced by personal experiences or subjective impressions, while the official data provides an accurate, long-term view of price trends.

1.6. Impact on Purchasing Power

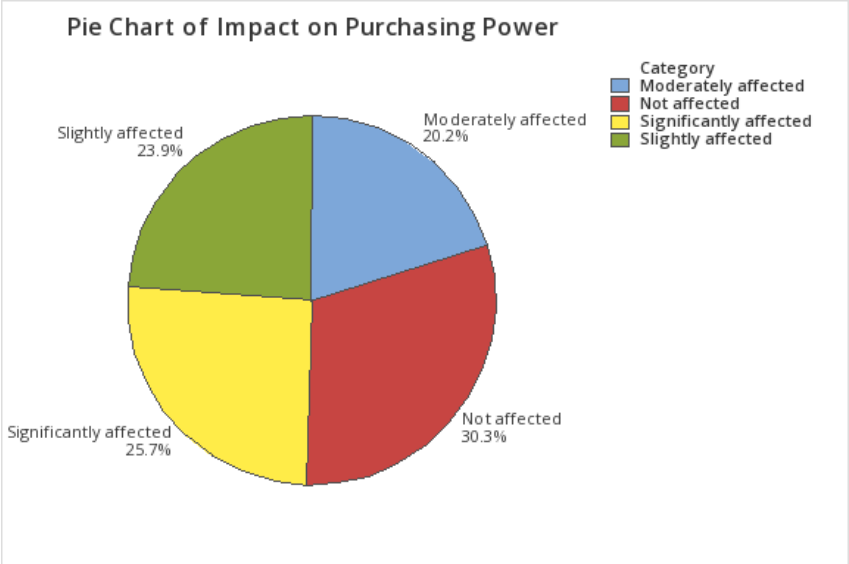

Figure 7: A pie chart showing the distribution of Impact on Purchasing Power

This pie chart illustrates the distribution of the impact of inflation on the purchasing power of individuals across four categories: "Not Affected," "Slightly Affected," "Moderately Affected," and "Significantly Affected." According to the chart, 30.3% of individuals report being "Not Affected" by inflation, which represents the largest proportion. Meanwhile, 25.7% fall under the "Significantly Affected" category, indicating a substantial share of people experiencing notable financial strain. 23.9% are "Slightly Affected," suggesting a minor but noticeable impact, and 20.2% are "Moderately Affected," representing those with a balanced level of impact between slight and significant. These percentages highlight the varying degrees of inflation's effect on the population, with a notable portion of the population experiencing mild to significant financial adjustments.

2. Relation Analysis

2.1 Monthly Income (SAR) vs Impact on Purchasing Power

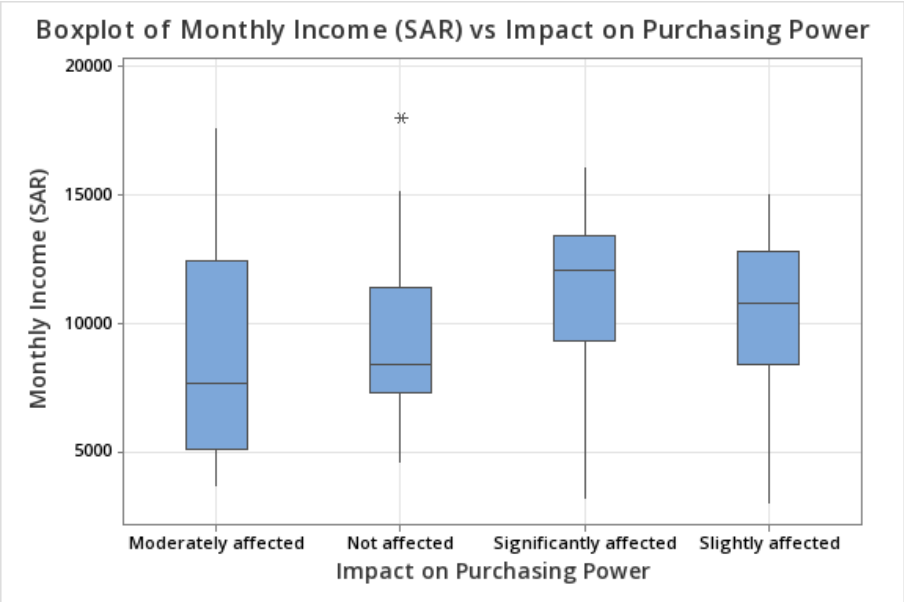

Figure 8: A side-by-side boxplot of Monthly Income (SAR) And Impact on Purchasing Power

The distribution of Monthly Income SAR by Impact on Purchasing Power is different. In the Moderately Affected group, the IQR is the widest, at 7,314.25 SAR, with a median of 7,719 SAR and incomes ranging from 3,733 SAR to 17,575 SAR, indicating significant variability. In contrast, the Significantly Affected group has an IQR of 4,114.3 SAR, a higher median of 12,106 SAR, and a range from 3,236 SAR to 16,086 SAR, reflecting a more consistent distribution among middle-to-higher income respondents. There are no outliers in either group, which indicates that the data is relatively concentrated within the expected boundaries. The overall analysis done shows that those affected moderately have greater income variation, while those greatly affected are mostly within higher-income brackets, though also very strongly burdened.

	Mean	Minimum	Q1	Median	Q3	Maximum	Outlier	IQRange
Moderately affected	**9179.68**	**3733**	**5149.25**	**7719**	**12463.5**	**17575**	**-**	**7,314.25**
Not affected	**9240.94**	**4629**	**7364**	**8426**	**11430**	**17995**	**17995**	**4,066**
Significantly affected	**11067.2**	**3236**	**9339.5**	**12106**	**13453.8**	**16086**	**-**	**4,114.3**
Slightly affected	**10257.1**	**3075**	**8433.5**	**10790.5**	**12800.8**	**14996**	**-**	**4,367.3**

Table 4:Descriptive analysis of the regularity in Monthly Income (SAR) vs Impact on Purchasing Power

2.2 Monthly Income (SAR) vs Job Type

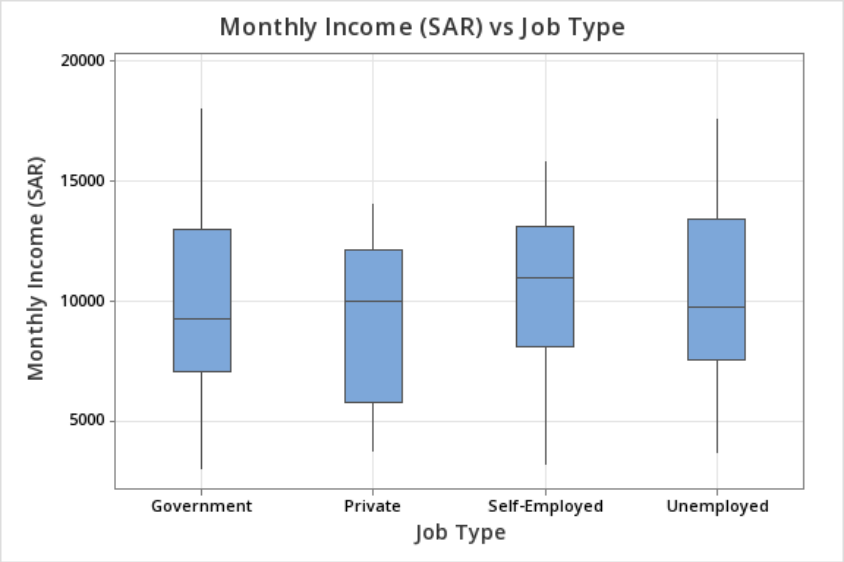

Figure 8: A side-by-side boxplot of Monthly Income (SAR) And Job Type

The government group jop has an average income of 9,790.14 sar and a median of 9,278 sar, with an iqr of 5,923.5 sar. Incomes in this group vary from 3,075 sar to 17,995 sar, showing moderate dispersion. The self-employed group jop presents a higher average income of 10,279.8 sar and a median of 11,012 sar, with a much larger iqr of 10,987 sar. Income in this group varies between 3,236 sar and 15,806 sar, so it is wider. No detected outliers have been seen within either group; thus, this supports the fact that their distribution ranges are relatively consistent.

	Mean	Minimum	Q1	Median	Q3	Maximum	Outlier	IQRange
Government	**9790.14**	**3075**	**7067.5**	**9278**	**12991**	**17995**	**-**	**5923.5**
Private	**9497.08**	**3780**	**5831**	**10014**	**12170.3**	**14069**	**-**	**6339.3**
Self-Employed	**10279.8**	**3236**	**8147**	**11012**	**13134**	**15806**	**-**	**10987**
Unemployed	**10188.0**	**3733**	**7552**	**9781**	**13432**	**17575**	**-**	**5880**

Table 5 : Descriptive analysis of the regularity in Monthly Income (SAR) vs Job Type

2.3 Increase In Living Costs (Sar) Vs Change In Purchasing Habits

INCREASE IN LIVING COSTS	CHANGE IN PURCHASING HABITS				All
	Moderate change	No change	Significant change	Slight change	
10-20%	12	6	5	8	31
Less than 10%	4	7	7	11	29
More than 20%	4	7	6	9	26
No increase	9	5	7	2	23
All	29	25	25	30	109

Table 6 :A 4x4 two-way table of INCREASE IN LIVING COSTS Vs CHANGE IN PURCHASING HABIT

Suspecting an association between Increase in Living Costs and Change in Purchasing Habits, the respondents were asked about the extent of cost increases they experienced and how their purchasing habits changed as a result. A two-way table representing the relationship between Increase in Living Costs and Change in Purchasing Habits is shown above. According to the table, among those who reported a cost increase of less than 10%, 13.79% experienced a moderate change, 24.14% reported no change, 24.14% experienced a significant change, and 37.93% experienced a slight change. Conversely, among those who reported no increase in costs, 39.13% experienced a moderate change, 21.74% reported no change, 30.43% experienced a significant change, and only 8.7% experienced a slight change. These differences in conditional distributions suggest a potential relationship between Increase in Living Costs and Change in Purchasing Habits, which warrants further investigation using a chi-square test to verify the association. The observed variation highlights the differing impacts of cost changes on consumer behavior.

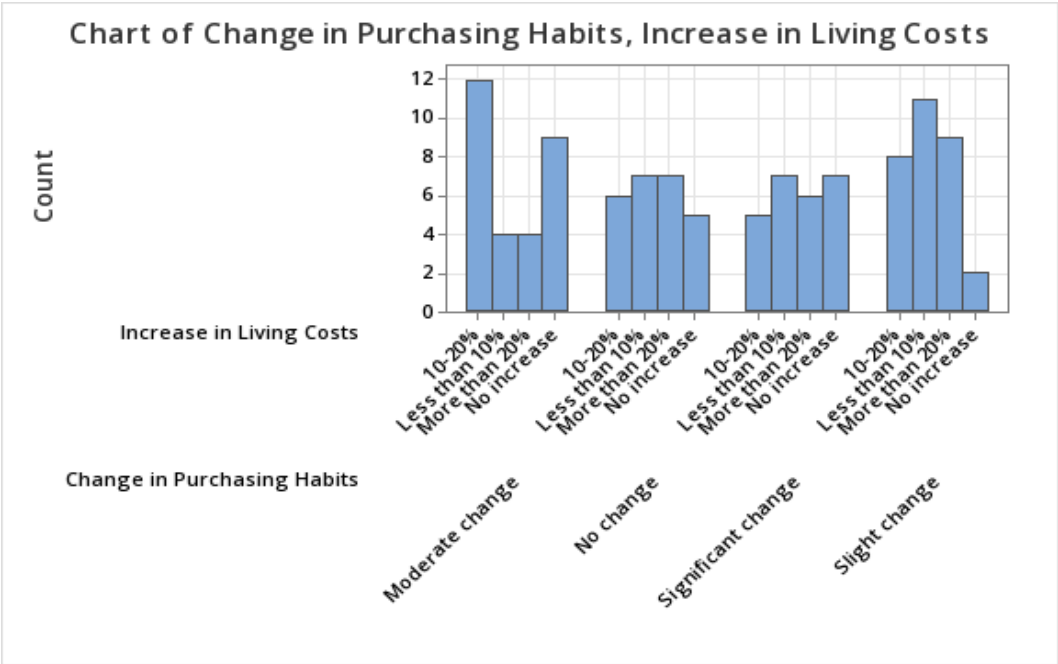

Figure 9: The conditional distribution of the relation between increase in living costs and change in purchasing habit

It is observable from the chart that the conditional distribution is not uniform across different levels of Increase in Living Costs and Change in Purchasing Habits. Thus, we can use the Chi-square test to assess the relationship between these two categorical variables. First, we establish the null hypothesis, H_0, and the alternative hypothesis, H_a, for the two categorical variables of one population:

- H_0: Increase in living costs has no relation to a change in the purchasing habit - there is no relationship (variables are independent).

- H_a: There exists a relationship between Increase in Living Costs and Change in Purchasing Habits, (the variables are dependent.)

2- Expected counts:

INCREASE IN LIVING COSTS	CHANGE IN PURCHASING HABITS				Total
	Moderate change	**No change**	**Significant change**	**Slight change**	
10-20%	8.248	7.110	7.110	8.532	31.0
Less than 10%	7.716	6.651	6.651	7.982	29.0
More than 20%	6.917	5.963	5.963	7.156	25.999
No increase	6.119	5.275	5.275	6.330	22.9
Total	29.0	24.9	24.9	30.0	108.99

Table 7:INCREASE IN LIVING COSTS Vs CHANGE IN PURCHASING HABIT

After calculating the expected counts, as shown in Table 2x2 , we observe that it is a multi-dimensional table where all expected counts are greater than 5. Furthermore, the average of expected counts is greater than 5, satisfying the requirements of the chi-square test. Therefore, we can perform the chi-square test:

$\chi 2 = 12.290$;

DF = 9 ;

P-value = 0.197

Referring to the chi-square distribution table, the critical value at ($\alpha = 0.05$) is higher than the calculated chi-square value, and the P-value (0.197 > 0.05). Since ($\alpha <$ P-value), we fail to reject the null hypothesis *H0* . This means the test is not statistically significant.

2.4 Number of Dependents and Essential Expenses (SAR)

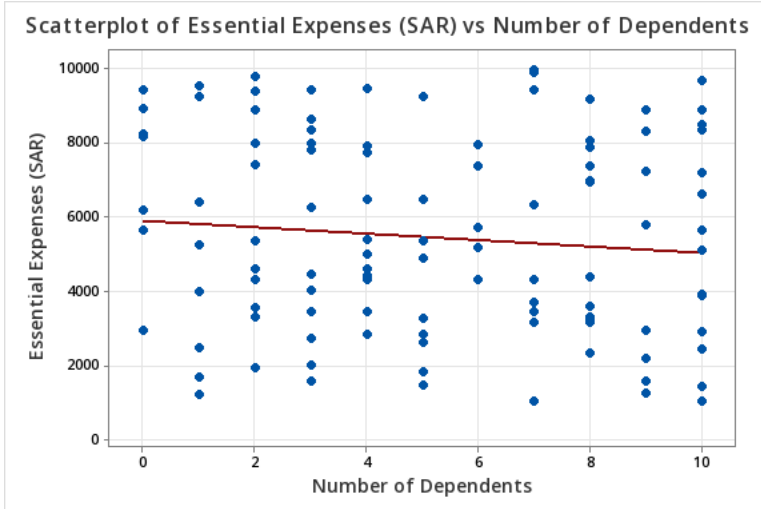

Figure 10: A scatterplot of Number of Dependents Vs Essential Expenses (SAR)

The regression equation = Essential Expenses (SAR) = 5931 - 88.45 Number of Dependents

The participants were asked about the number of dependents to explore its association with essential expenses (SAR), as dependents might potentially influence a household's financial needs. A scatterplot of Number of Dependents and Essential Expenses (SAR) is shown above. It is evident that there is a weak, negative, linear relationship between the number of dependents and essential expenses. The correlation coefficient is $r=-0.182$, indicating that approximately 3.3% of the variation in essential expenses (response variable) can be explained by the number of dependents (explanatory variable).

The scatterplot suggests that the relationship between the two variables is not strong, and several outliers are visible. One potential outlier is an individual with 10 dependents who reported very low essential expenses. Additionally, Minitab software identifies other influential points that slightly deviate from the overall trend.

The hypotheses for this analysis are as follows:

- **H_0 (Null Hypothesis):** There is no relationship between the number of dependents and essential expenses ($\beta 1=0$).
- **H_a (Alternative Hypothesis):** There is a negative relationship between the number of dependents and essential expenses ($\beta 1<0$).

▪ Regression inference conditions:

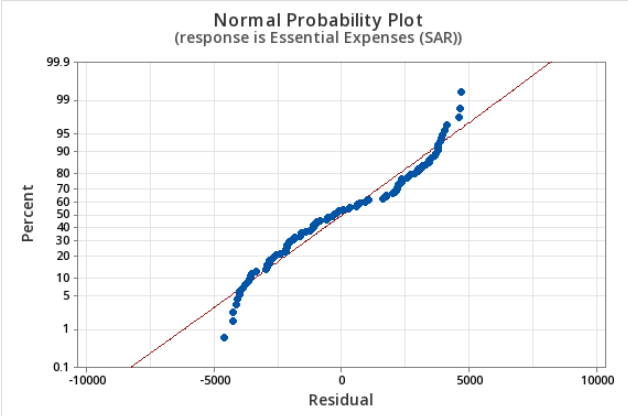

Figure 11: The normal probability plot of the residuals for Number of Dependents

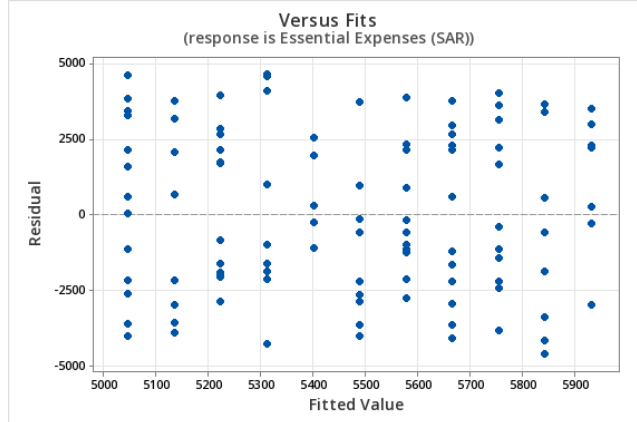

Figure 12: A residual plot of Number of Dependents

The normal probability plot in Figure 10 exhibits some minor deviations from a straight line, indicating that the data are not perfectly Normally distributed for each value of x. However, the sample size is large in this study, n = 109, and regression procedures are robust to minor violations of Normality. Additionally, the residuals plot exhibits a fair scatter of residuals around the horizontal line at 0 throughout the range of fitted values, as depicted in Figure 11. This is an indication that there are no significant patterns or heteroscedasticity in the residuals. Since all assumptions underlying the regression analysis are met, inference procedures for the regression model can be applied accordingly.

Significance test for the slope:

Term	Coef	SE Coef	T-Value	P-Value
Constant	5931	495	11.98	0.000
Number of Dependents	-88.5	81.0	-1.09	0.277

Table 8:Significance test for the slope of the relation between Number of Dependents and Essential Expenses (SAR)

The 95% confidence interval for the slope is:

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	**1**	**8569311**	**8569311**	**1.19**	**0.277**
Error	**107**	**768124368**	**7178732**		
Total	**108**	**776693679**			

Table 9: The ANOVA table for the relation between Number of Dependents and Essential Expenses (SAR)

According to the information given in Table 8 and 9, the P-value for the two-sided test is P = 0.277 and for the one-sided test is P = 0.1385 which is less than our significance level $\alpha = 0.05$. As a result, We have enough evidence to reject the null hypothesis H0 which denies the relationship between the number of dependents and essential expenses, and accept our alternative hypothesis Ha . In other words, the test is statistically significant, and a conclusion can be drawn that there is a negative linear relationship between the number of dependents and essential expenses. As the number of dependents increases, the essential expenses tend to slightly decrease.

Conclusions

In essence, through this study, it is observed that inflation significantly affects the purchasing power of individuals in Saudi Arabia. A number of insights were deduced from the analysis concerning the relationship between key factors such as monthly income, number of dependents, essential expenses, and changes in purchasing habits. The following are the key findings:

- Impact of Inflation on Purchasing Power:

The study found that as inflation increases, the purchasing power of individuals is significantly affected, with higher-income groups showing resilience compared to lower-income groups.

The majority of participants reported moderate to significant changes in their purchasing habits as living costs increased.

- Relationship Between Number of Dependents and Essential Expenses:

There was a weak negative linear relationship between the number of dependents and essential expenses. This may imply that households with more dependents tend to manage their essential expenses more conservatively.

- Chi-Square Analysis:

The chi-square test for the association of increases in the cost of living and changes in purchasing habits was not significant. It would appear that a change in purchasing habits may not be based on inflation but rather other exogenous factors.

- Regression Analysis:

Regression analysis of essential expenses indicated a minor but perceptible pattern whereby the number of dependents tends to decrease spending. While statistically weak, this reflects changes in financial management with larger families.

- Other Observations:

There was a significant difference in the amount of savings used. Whereas 44% of respondents did not use their savings, a substantial percentage claimed to have utilized between 5,000 and 10,000 SAR to manage the increase in expenses.Fuel and transportation costs were variable, with most respondents showing moderate expenses but a small proportion reporting much higher costs.

- Recommendations for Future Studies

3 Sample Size Expansion: A larger sample size would provide more solid results and represent diversity in the population.
4 Including More Variables: Variables such as level of education, job stability, and regional differences can give more insight into the impact of inflation.

5 Policy Implications: The findings point out that there is a dire need for focused policies on the support of low-income families and dampening inflation, especially in key sectors like health and education.

Despite the valuable insights provided by this study, certain limitations exist, such as the relatively small sample size and the exclusion of key variables like education level and job stability, which could influence the results. Future studies should aim to expand the sample size and include additional variables to gain a more comprehensive understanding of inflation's effects on different social groups.

The study recommends the development of targeted economic policies to support those most affected by inflation, such as financial aid programs and subsidies for essential goods. Efforts should also focus on price stabilization in critical sectors such as housing and food. Implementing these recommendations will help mitigate the adverse effects of inflation and ensure the achievement of Vision 2030 goals, fostering economic equity and well-being for all.

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Statistical Concepts

Alternative Hypothesis (Ha): A hypothesis that proposes there is an effect or relationship. For example, there is a negative relationship between inflation rates and purchasing power. Evidence is gathered to support this hypothesis.

ANOVA Table: Refers to the Analysis of Variance table. It summarizes the components of variance in data, including the sum of squares (SS), degrees of freedom (DF), mean squares (MS), F-statistic, and P-value.

Categorical Variable: A variable with values that represent categories or groups, such as job type or place of residence.

Chi-Square Test: A statistical test that checks whether observed frequencies in categories differ from expected frequencies. It is used for categorical data.

Confidence Interval (CI): A range of values that likely contains the true parameter value, given a specified confidence level (e.g., 95%).

Correlation: Denoted as *r*, it measures the strength and direction of the linear relationship between two variables.

Dependent (Response) Variable: The variable being measured or predicted in a study, such as purchasing power.

Independent (Explanatory) Variable: The variable that explains changes in the response variable, such as inflation rates.

Mean: The average value of a data set, calculated by dividing the sum of all values by the number of observations**.**

Median: The middle value in an ordered data set, where half the values are less than the median and half are greater.

Mode: The most frequently occurring value in a data set.

Null Hypothesis (H0): A hypothesis that assumes no effect or relationship exists. For example, there is no relationship between inflation rates and purchasing power.

One-Sided Test: A test that evaluates a directional hypothesis, focusing on one end of the distribution.

Outlier: A data point that lies significantly outside the range of other observations.

P-Value: The probability of observing a result as extreme as the one obtained, assuming the null hypothesis is true.

Significance Level (α): A threshold (commonly 0.05) against which the P-value is compared to decide whether to reject the null hypothesis.

Skewed Distribution: A distribution that is not symmetrical, where one tail is longer than the other.

Two-Sided Test: A test that evaluates the possibility of an effect in either direction, producing a two-sided P-value.

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